

USI, Risk Management, Spring 2025

Case #1

Lecturer: **Alberto Plazzi** TA: **Marco Zanotti**

Deadline: April 9, 2025 (23:59)

It's closing of March 13, 2025 and you are the risk manager of *USI Capital Management*, a hedge fund that invests primarily in mid and large-cap Swiss companies. The fund has currently the following positions on 3 **equities** trading on the Swiss Stock Exchange: *Nestlé* (NESN), a food and drink processing conglomerate, *Swiss Re* (SREN), the world's largest reinsurance company, and *Novartis* (NOVN), a pharmaceutical corporation.

Time-series of the prices for the 3 stocks and the returns of the Swiss Market Index (SMI) can be found in the spreadsheet 'Data' of **Data_Case_1.xlsx**. Use whenever needed simple returns and sample variances and covariances (.S in Excel):

Part I: First, let's assume that the portfolio was formed on **closing of November 21, 2022** and we want to assess its riskiness by computing the fund's **VaR(99%,1)** *each day until March 13, 2025*. The portfolio consists of a short position of 30'000 shares in NESN, a long position of 70'000 shares in SREN and a long position of 80'000 shares in NOVN.

1. Obtain VaR(99%,1) of each equity position separately assuming normally distributed returns. For this, calculate volatilities on November 21, 2022 based on past 125 returns and keep them constant throughout the period. Re-compute the same VaR(99%, 1) using the past 250 returns. Briefly comment on the efficacy of this approach.
2. Calculate the fund's VaR(99%,1) using an Asset-Normal approach. For this, compute correlations on November 21, 2022 based on past 250 returns and keep them constant throughout the period. Notice that the **number of shares** (not the value!) of each position remains fixed throughout the period. This is known as a buy-and-hold strategy. Compare the resulting VaR with the sum of individual VaRs (a.k.a. 'undiversified VaR'). What do you conclude?
3. Compute the percentage of days where the actual loss exceeded the AN VaR(99%,1), is the result in line with what you expected?
Now recalculate the same percentage only for 2024 (starting January 3, 2024). Is the result in line with what you expected?
Briefly comment on this.
4. For each single position at each day compute its CVaR and RCVaR. Plot the RCVaRs over time and comment.

5. Due to incoming flows in the portfolio, you plan the following trades for tomorrow (March 14, 2025): buy 2'000'000 CHF in stock NESN, buy 1'000'000 CHF in SREN, sell 1'000'000 CHF in NOVN. Based on today's Asset-Normal VaR estimates, use the incremental VaR to determine what would be the approximate VaR after the trade. Compare it to the VaR recomputed using the new holdings.

Part II: Now, your line manager wants to use a different strategy, which considers both a mapping of the fund's equity exposures to the Swiss Market factor as well as time-variation in volatility, to finally compute a Sharpe Diagonal VaR(95%,1) for March 14, 2025 (**so, just tomorrow's VaR (not a time series) needs to be computed for this part and the following.**

Moreover, assume the trade in point 4. did NOT take place, i.e. the holdings are still: a short position of 30'000 shares in NESN, a long position of 70'000 shares in SREN and a long position of 80'000 shares in NOVN.).

6. Compute the stocks' betas with respect to the SMI Market factor based on the full sample. Which of the stocks have the highest/lowest systematic risk?
7. You now want to account for time-varying volatility of the SMI Market factor with a RiskMetrics model, i.e. a EWMA model with $\lambda = 0.94$. To this end, use data starting from Nov. 12, 2021 onwards and start the series (and the updating scheme) from the first squared return (i.e. set $\sigma_0^2 = r_{11thNov2017}^2$). Plot the resulting series of conditional variances and comment on it. Use the model to estimate the conditional variance of March 14, 2024 SMI's return. Redo the exercise but instead of RiskMetrics use a GARCH(1,1) model on the SMI return.
8. Compute the Sharpe Diagonal VaR(99%;1) for the portfolio, using the predicted volatilities i) from the RiskMetrics model and ii) from the GARCH(1,1) model. How are they different from that computed in point 2) above for the same date? Which of the three models do you trust more? Comment.